

PLC Automation Project

TIA Portal + Factory I/O

Project Documentation



Complete simulated production line in Factory I/O.

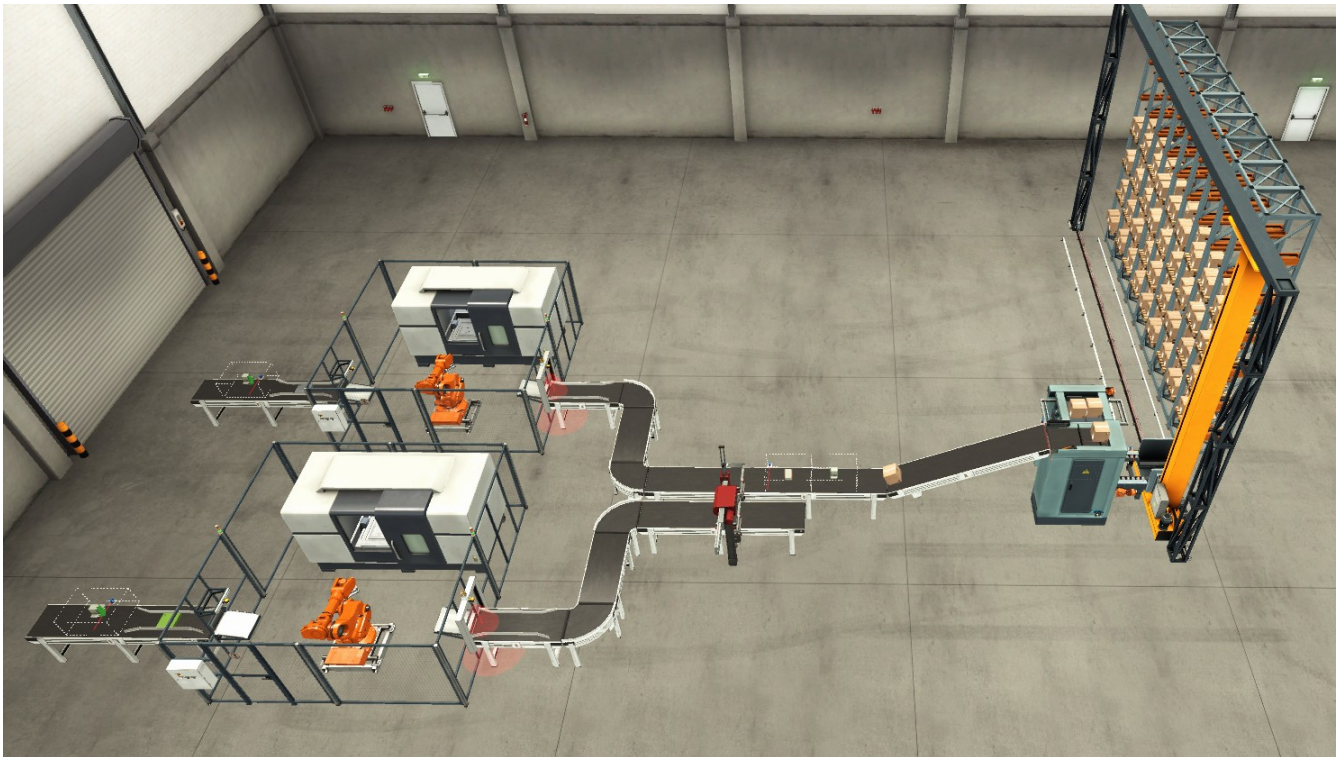
SIMATIC S7-1500 | PLC program | KTP900 Basic PN HMI

1. Project overview

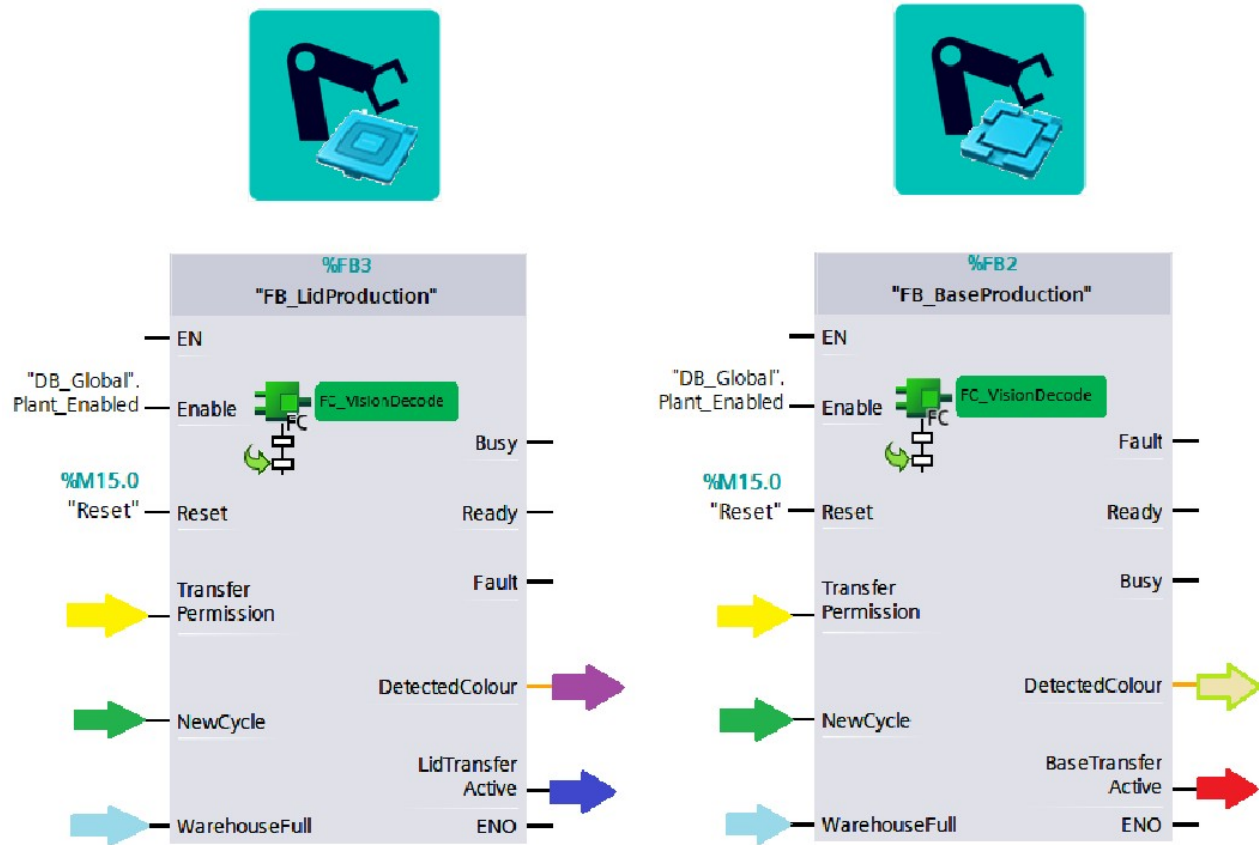
This project shows a complete simulated production line. Two stations make two parts of one product: a Lid and a Base. The parts are checked by vision sensors and then moved to the assembly station. After assembly, the finished product is packed, placed on a pallet and stored automatically in the warehouse.

The PLC program is split into several blocks. Each block is responsible for one part of the machine. OB1 mainly calls these blocks and connects the most important signals between them.

- FB3 - Lid production
- FB2 - Base production
- FB4 - Assembly sequence
- FB7 - Assembly actuators
- FB5 - Palletizer
- FB6 - Warehouse
- FB8 - Product tracking
- FC1 - Vision sensor evaluation
- FC3 - HMI information
- FB1 - PLC and Factory I/O reset



2. Production of Lid and Base



The picture shows the main inputs and outputs of FB_LidProduction and FB_BaseProduction. Only the signals needed to understand the connection between the blocks are shown.

The Lid and Base blocks work in a very similar way. A new part is created, transported into the machine, processed and checked by a vision sensor. The finished part waits until the assembly station is ready to receive it.

Plant_Enabled - allows the production process to run.

Reset - resets the production block.

TransferPermission - tells the production block that the assembly station is ready for the finished part.

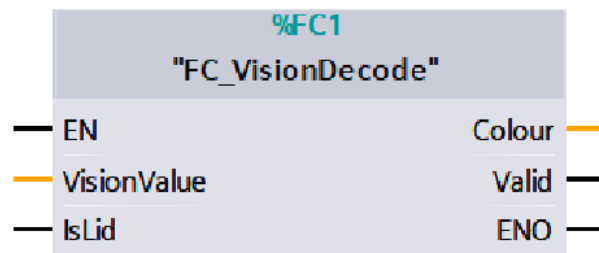
NewCycle - tells the production block that the previous product has been assembled and a new cycle can start.

WarehouseFull - stops a new production cycle when the warehouse is full.

DetectedColour - stores the detected colour of the finished part.

LidTransferActive / BaseTransferActive - show when a finished Lid or Base is being moved to the assembly station.

3. FC_VisionDecode



FC1 is used by both production blocks to read the vision sensor values.

FC_VisionDecode is a small function used inside the Lid and Base production blocks. It checks the value from the vision sensor and converts it into a simple colour number.

VisionValue (IN) - value received from the Factory I/O vision sensor.

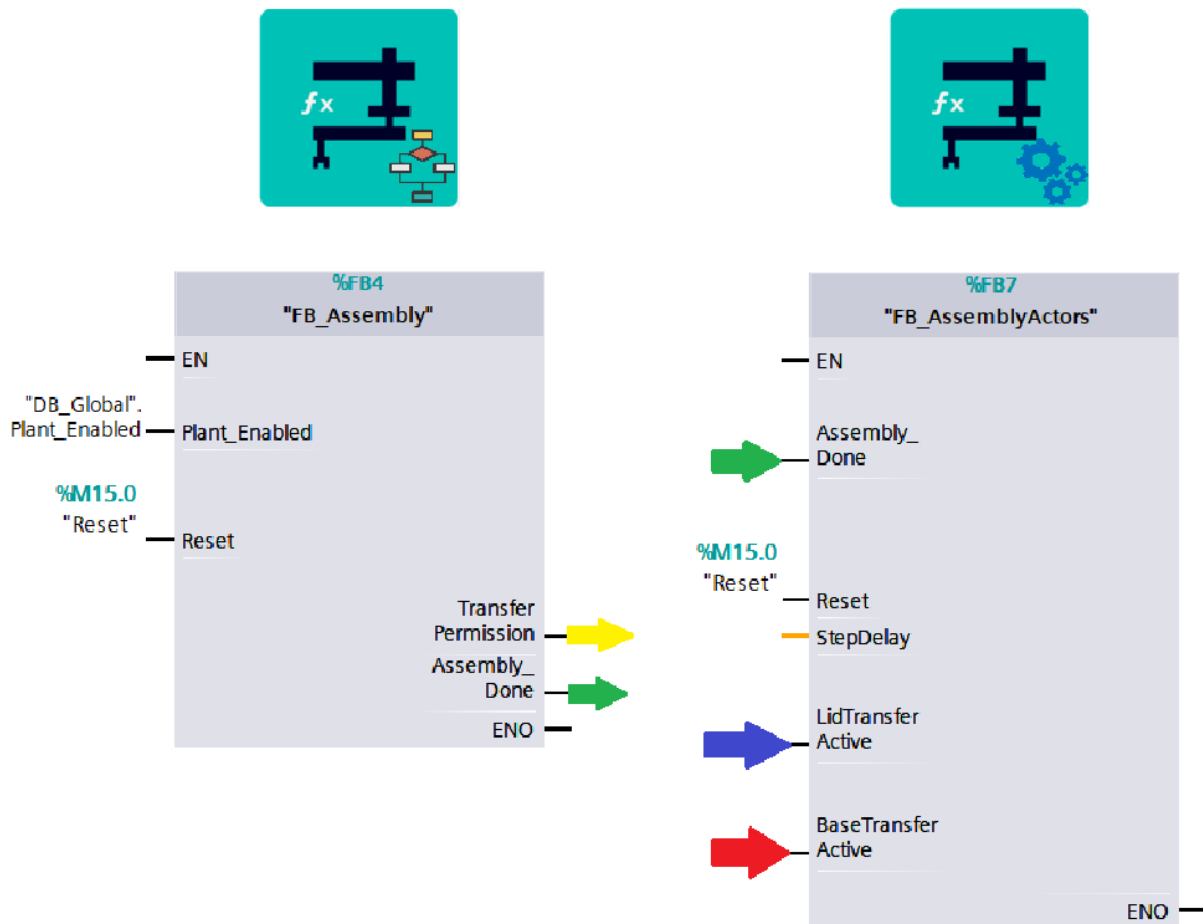
IsLid (IN) - selects whether the value belongs to a Lid or a Base.

Colour (OUT) - gives the detected colour: 1 = Blue, 2 = Green, 3 = Metal.

Valid (OUT) - becomes TRUE when the sensor value is one of the expected values.

The Lid and Base use different sensor values. FC1 converts these values into the same colour numbers, so the rest of the program only works with Blue, Green and Metal.

4. Assembly



FB_Assembly controls the assembly steps. FB_AssemblyActors controls the movements of the assembly station.

The assembly part is split into two blocks. FB_Assembly decides which step is active and when the station is ready. FB_AssemblyActors then controls the conveyors, press, gripper, remover and package emitter.

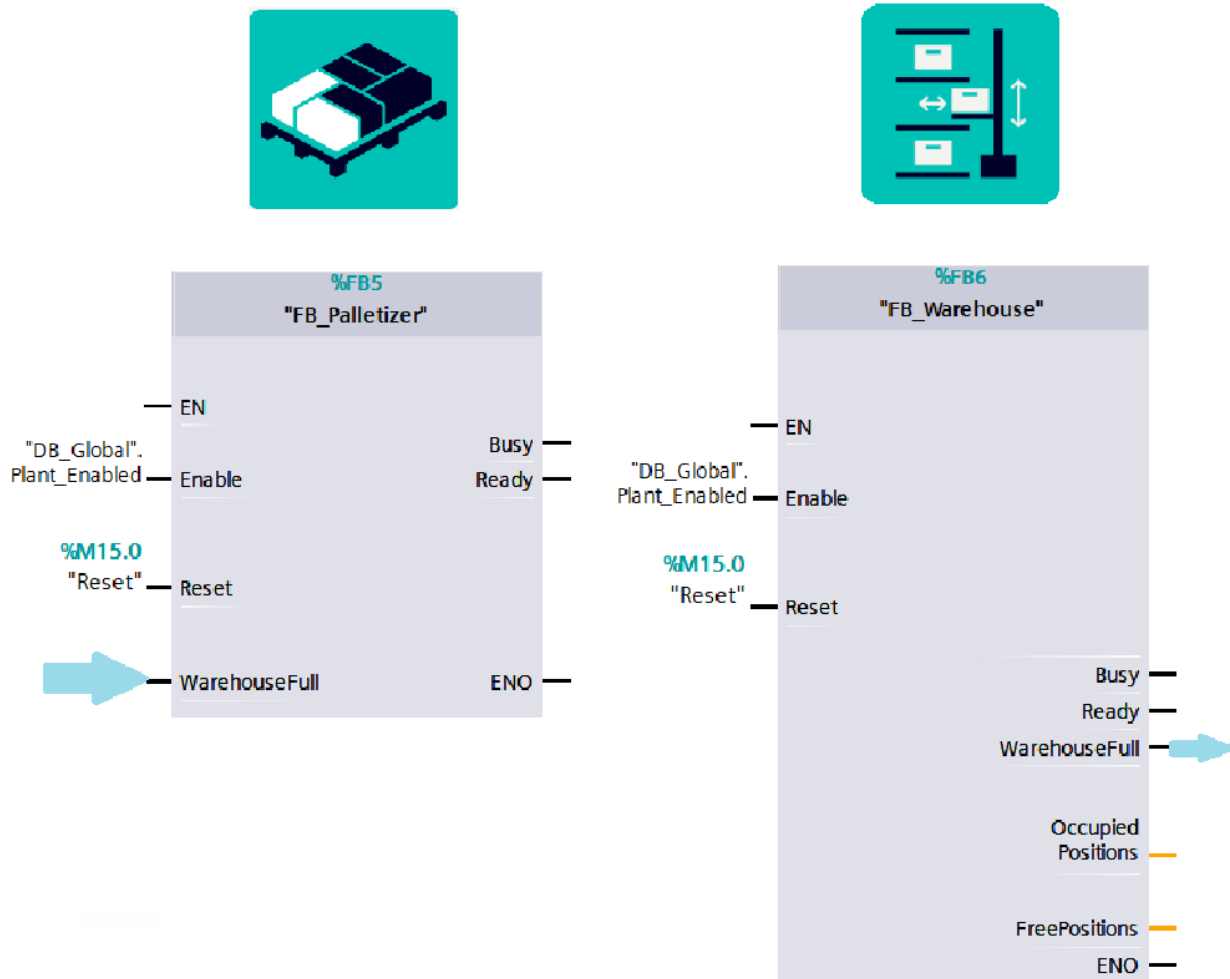
TransferPermission - tells the Lid and Base production blocks that the assembly station is ready.

Assembly_Done - becomes active when one complete Lid + Base product has been assembled. It is also used for product counting.

LidTransferActive / BaseTransferActive - tell the assembly actuators which part is currently being transferred.

The internal Step_0 to Step_8 signals are used inside the assembly program. They are not all shown in the simplified diagram because they are only needed for the internal sequence.

5. Palletizer and warehouse



Main blocks used after assembly: FB_Palletizer and FB_Warehouse.

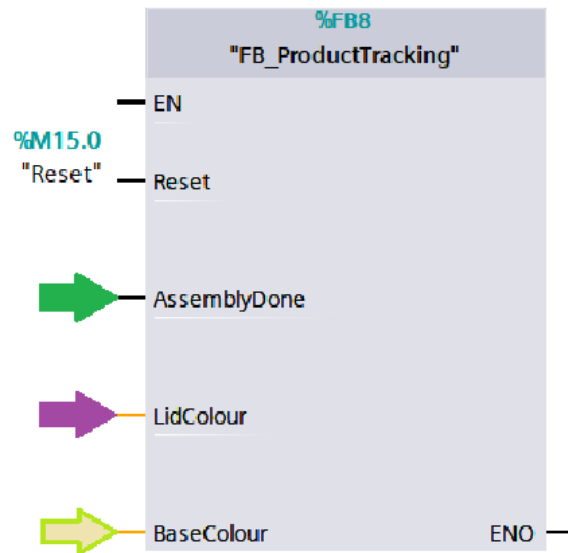
After assembly, every finished product becomes one package. The palletizer collects the packages. Two packages are moved at a time, and after two movements the pallet contains four packages.

The warehouse receives the full pallet and controls the stacker crane. The warehouse has 54 storage positions. The HMI shows how many positions are occupied and how many are free.

WarehouseFull - becomes active when there is no free storage position. In this case, the production line stops creating new material.

Busy / Ready - show whether the palletizer or warehouse is working or waiting.

6. Product counting



FB8 counts the nine possible Lid/Base colour combinations.

FB_ProductTracking is used only for counting finished products. When Assembly_Done changes to TRUE, one finished product is counted using the detected Lid and Base colours.

There are three Lid colours and three Base colours, so there are nine possible combinations. The values are shown on the HMI.

7. DB_Global

DB_Global stores values that are used by several parts of the program or by the HMI. It is mainly used to keep important commands and display values in one place.

Plant_Enabled - shows whether the production line is started or stopped.

HMI_LidStatus / HMI_BaseStatus - values used to show the current Lid and Base status on the HMI.

HMI_WarehouseStatus - value used to show the warehouse status on the HMI.

HMI_OccupiedPositions / HMI_FreePositions - number of occupied and free warehouse positions.

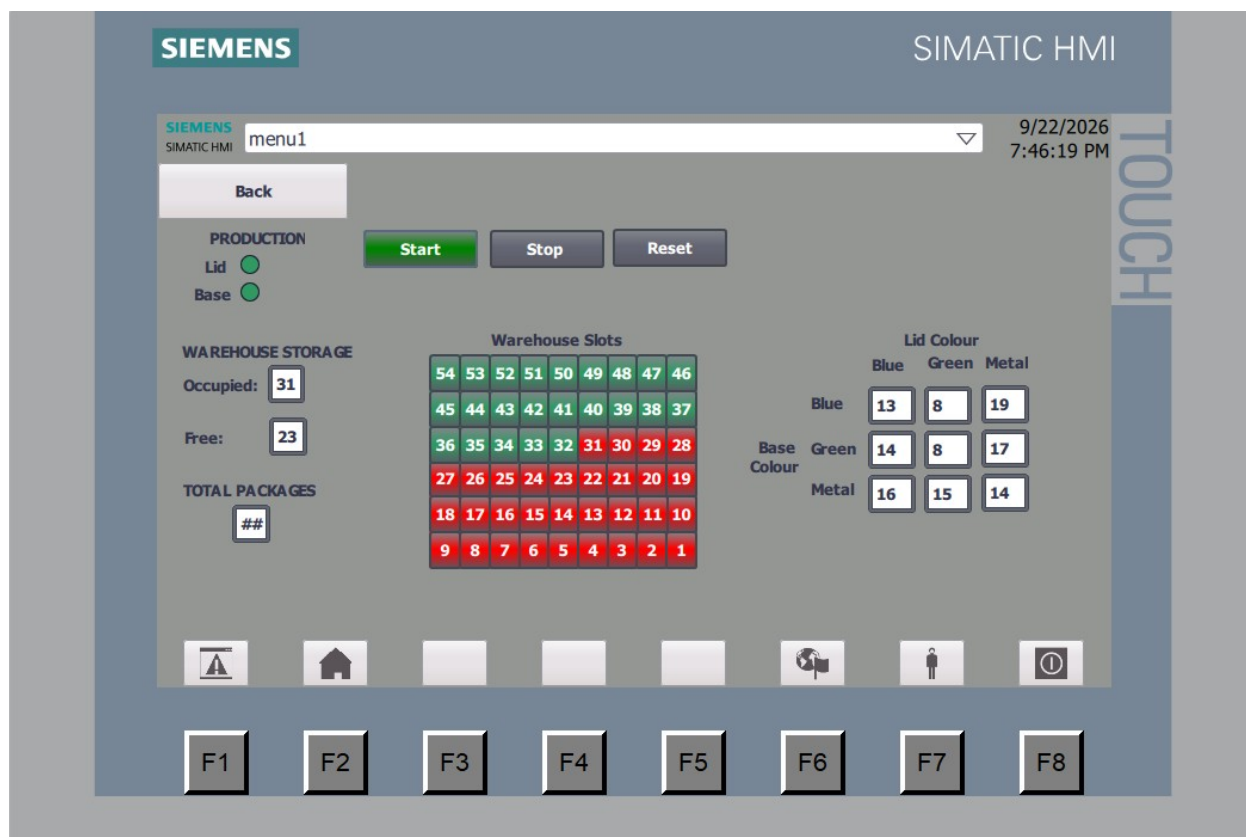
HMI_TotalPackages - total number of packages stored in the warehouse.

Warehouse_Full - value used by the HMI to show that the warehouse is full.

HMI_AlarmWord - value used to show the Warehouse Full alarm on the HMI.

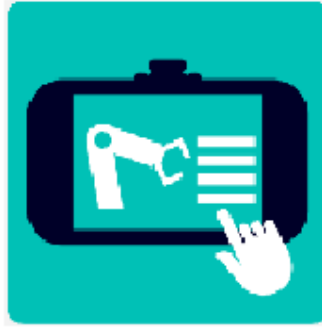
HMI_AckWord - value used when the operator confirms the alarm or requests a reset.

8. HMI



Main KTP900 Basic PN screen.

The HMI gives the operator the main controls and information. It includes Start, Stop and Reset, Lid and Base status, occupied and free warehouse positions, the total package count and the nine product colour combinations.



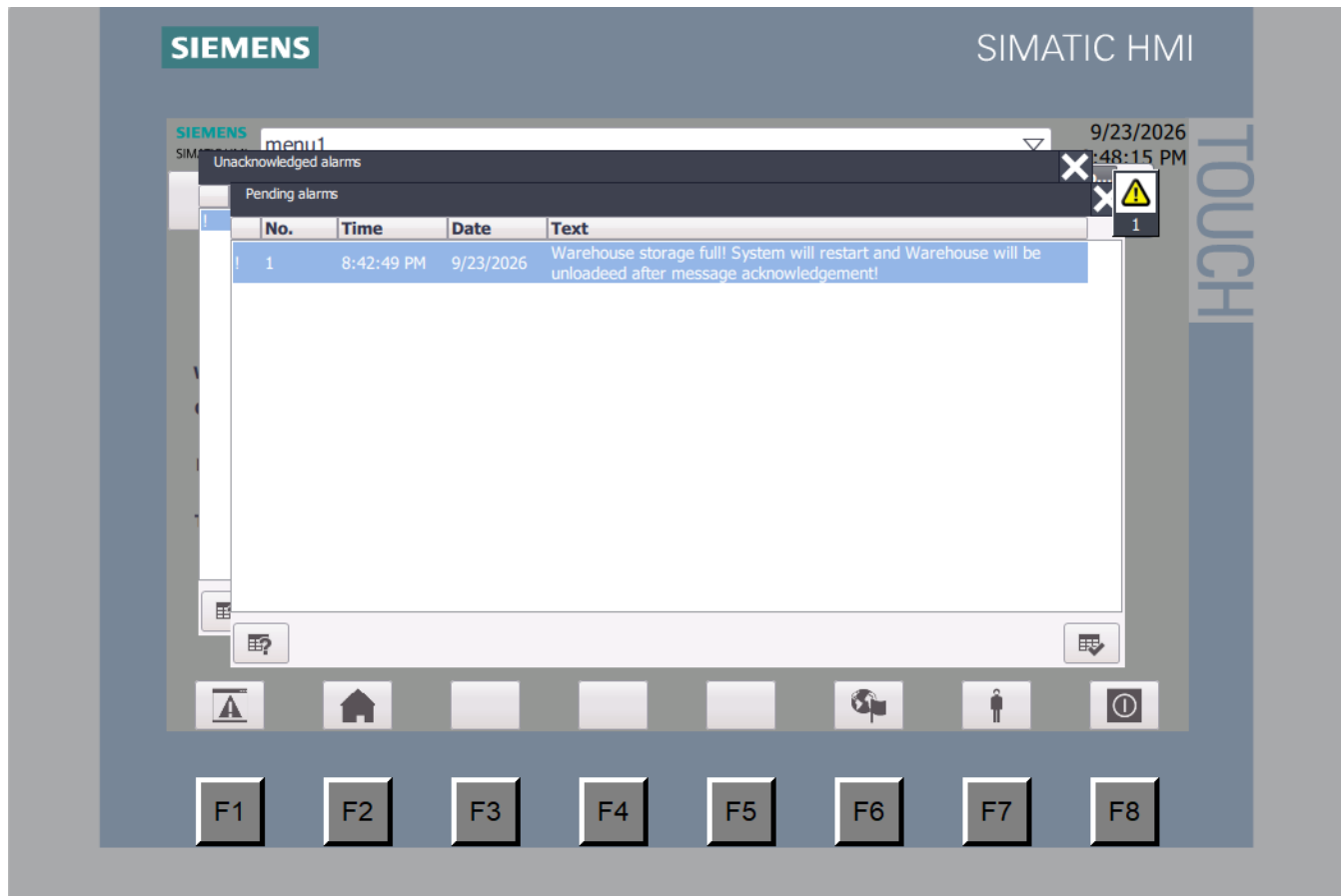
FC3 prepares the values that are shown on the HMI.

FC_HMI_Status reads important values from the PLC blocks and converts them into simple status numbers for the HMI.

- Lid/Base: 0 = Idle, 1 = Ready, 2 = Busy.
- Warehouse: 0 = Idle, 1 = Ready, 2 = Busy, 3 = Warehouse Full.
- HMI_TotalPackages = occupied warehouse positions $\times$ 4, because one pallet contains four packages.

Ret_Val is the return value of FC3. It is set to TRUE when the function finishes normally. It is not used by another block, so it is not important for the process.

9. Warehouse Full alarm

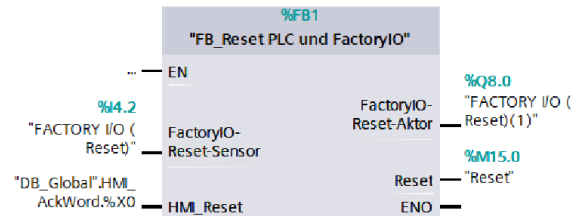


Warehouse Full alarm on the HMI. The operator has to confirm the message.

When the warehouse is full, the PLC sends this information to the HMI. The HMI then shows a Warehouse Full alarm message.

After the operator confirms the message, the HMI sends a reset request back to the PLC reset block.

10. Reset



FB1 combines the Factory I/O reset and the HMI reset into one common reset signal.

The reset block creates one reset signal for the complete project. This makes sure that all main blocks are reset in the same way.

FactoryIO-Reset-Sensor (IN) - Factory I/O reset input %I4.2.

HMI_Reset (IN) - reset request from the HMI.

FactoryIO-Reset-Aktor (OUT) - Factory I/O reset output %Q8.0.

Reset (OUT) - common PLC reset %M15.0.

The common reset signal is connected to Lid Production, Base Production, Assembly, Assembly Actuators, Palletizer, Warehouse and Product Tracking.

11. Main signal connections

The most important connections between the blocks are:

- DB_Global.Plant_Enabled -> starts or stops the main process blocks.
- FB_Assembly.TransferPermission -> tells Lid and Base Production when Assembly is ready.
- FB_Assembly.Assembly_Done -> tells the production blocks that a new cycle can start and tells Product Tracking to count one product.
- LidTransferActive / BaseTransferActive -> tell the Assembly Actuators which part is being transferred.
- DetectedColour -> sends the detected Lid and Base colours to Product Tracking.
- FB_Warehouse.WarehouseFull -> stops new production when the warehouse has no free position.
- Factory I/O Reset or HMI reset -> FB_Reset -> common reset %M15.0.

12. Project summary

The project is divided into separate blocks for production, assembly, palletizing, warehouse, product counting, HMI information and reset. The detailed sensors, timers and step sequences remain inside the blocks that use them.

In simple terms: the production blocks make and check the parts, the assembly block joins them, the palletizer collects the packages, the warehouse stores them, the product tracking block counts them, the HMI shows the main information, and the reset block resets the complete system.